

Ruichen Xu

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SUMMARY

Ph.D. candidate in Applied Mathematics & Statistics at Stony Brook University. Research interests: **LLMs/agents for AI4Science**, **generative AI** (diffusion) for physics-/constraint-aware sampling, and **neural operators** for partial observations and inverse problems. I build reproducible pipelines and run large-scale experiments on SLURM/CUDA HPC clusters.

EDUCATION

- **Stony Brook University (The State University of New York)** Aug 2022 – Present (Expected 2027)
Ph.D., Computational Applied Mathematics (GPA 4.00/4.00) Stony Brook, NY, USA
- **Courant Institute of Mathematical Sciences, New York University (NYU)** Sep 2020 – May 2022
M.S., Mathematics (GPA 4.00/4.00) New York, NY, USA
- **University of California, Davis** Sep 2019 – Jun 2020
M.S., Statistics (GPA 3.94/4.00) Davis, CA, USA
- **Beijing University of Chemical Technology (BUCT)** Sep 2015 – Jun 2019
Bachelor, Financial Mathematics (89.33/100); Minor: Commercial Management Beijing, China

PROFESSIONAL EXPERIENCE

- **AI3, Stony Brook University** Jun 2026 – Jul 2026
REU Research Mentor Stony Brook, NY, USA
 - Mentors REU students on AI4Science research projects at AI3, including project scoping, literature review, experiment design, and technical implementation.
 - Guides students on reproducible research workflows, code organization, experiment tracking, and presentation preparation.

PROJECTS

- **JEPA for PDEs and Graph-Structured Scientific Data** Jan 2026 – Present
Tools: Python, PyTorch; Topics: JEPA, PDE dynamics, graph learning, self-supervised learning, AI4Science
 - Develops joint-embedding predictive architectures for PDE systems and graph-structured scientific data, targeting data-efficient latent physical representation learning under partial observations.
 - Designs modality-aware context/target prediction pipelines for PDE trajectories and scientific graphs, separating latent-space prediction from direct field or node reconstruction.
 - Studies transfer across grid-, mesh-, and graph-based physical systems with ablations on masking strategy, encoder depth, latent dimension, and prediction horizon.
 - **JEPA-PDE**: manuscript under review. **JEPA-Graph**: planned submission to *KDD 2026*; emphasizes scalable encoders, predictive objectives, and benchmark evaluation.
- **PartialObs-PDEBench & AI4Physics: Partial-Observation Benchmark + AI4PDE/AI4SDE Collection** 2025 – Present
Tools: Python; Topics: partial observations, benchmarking, diffusion, solver recommendation 
 - Maintains **AI4Physics**: curated AI4PDE/AI4SDE collection (taxonomy + tags + takeaways) and hosts ongoing results.
 - Develops PartialObs-PDEBench for evaluating PDE learning under sparse, irregular, sensor-style, and masked observation regimes.
 - Builds diffusion-based conditional samplers (APOD/APOD++) for reconstructing and forecasting PDE states from partial observations with unified metrics.
 - Organizes benchmark metadata, reproducible configs, and evaluation scripts to compare neural operators, diffusion models, and physics-aware baselines.
- **ORACLE: LLM-Guided Edits + Simulated Annealing for Structure-Based Drug Design (SBDD)** 2025 – Present
Tools: Python, PyTorch; Topics: LLM-guided molecular editing, multi-objective optimization
 - Combines LLM-proposed chemistry-aware edits with simulated annealing to explore local molecular neighborhoods while enforcing validity constraints.
 - Optimizes multi-objective scores including docking, QED, SA-score, diversity, and constraint satisfaction, with configurable objective weights.
 - Implements ablation and tracking pipelines for edit operators, temperature schedules, acceptance rules, and Pareto trade-off analysis.
 - *KDD 2026 submission in preparation*; emphasizes controllable molecular optimization, optimization curves, and chemically valid generation.

PUBLICATIONS

Peer-Reviewed – Conferences

- [c.1] **IAGA: Identity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** Jingxiang Qu, Wenhan Gao, Ruichen Xu, Yi Liu. *ICLR 2026 (Accepted)*. ([OpenReview](#))
- [c.2] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** Ruichen Xu, Claire Yu, Zongyu Wu, Siyao Wang, Luoyao Chen, Georgios Kementzidis, Haochun Wang, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#))
- [c.3] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** Wenhan Gao, Ruichen Xu, Yuefan Deng, Yi Liu. *ICLR 2025*. ([OpenReview](#)) ([ICLR Page](#))
- [c.4] **Kolmogorov–Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, R. Xu[†], L. Chen, G. Kementzidis, S. Wang, Y. Deng. *IJCNN 2025*. ([IEEE](#)); [†]co-first author; project lead.
- [c.5] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** Maximilian Cederholm, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).
- [c.6] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** Albert Ding, Siyao Wang, Haochun Wang, Ruichen Xu, Yuefan Deng. *NYSDS 2025, SIAM Proceedings*. ([SIAM DOI](#)); co-corresponding author (with Y. Deng).

Peer-Reviewed – Journals

- [j.1] **Multistep Generative Backmapping of Coarse-Grained Structures.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *Computer Physics Communications*, 327, 110286 (2026). ([DOI](#)) ([Elsevier Share Link](#))
- [j.2] **Dynamic Schwartz–Fourier Neural Operator for Enhanced Expressive Power.** Wenhan Gao, Jian Luo, Ruichen Xu, Yi Liu. *TMLR 2025*. ([OpenReview](#))
- [j.3] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** Wenhan Gao, Ruichen Xu, Haochun Wang, Yi Liu. *TMLR 2024*. ([OpenReview](#))

Peer-Reviewed Workshops

- [w.1] **APOD: Adaptive PDE-observation diffusion for physics-constrained sampling.** Ruichen Xu, Haochun Wang, Georgios Kementzidis, Chenhao Si, Yuefan Deng. *ICML 2025 Workshop on Assessing World Models (Poster)*. ([OpenReview](#))
- [w.2] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** Ruichen Xu, Kai Li, Haochun Wang, Georgios Kementzidis, Wei Zhu, Yuefan Deng. *2nd AI for Math Workshop @ ICML 2025 (Poster)*. ([OpenReview](#))
- [w.3] **An Iterative Framework for Generative Backmapping of Coarse-Grained Proteins.** Georgios Kementzidis, Erin Wong, John Nicholson, Ruichen Xu, Yuefan Deng. *ICML 2025 GenBio Workshop (Poster)*. ([OpenReview](#))

TEACHING EXPERIENCE

- **Stony Brook University – Department of Applied Mathematics & Statistics (AMS)** Aug 2022 – Present
Teaching Assistant & Instructor Stony Brook, NY, USA
 - **Teaching Assistant (Aug 2022 – May 2025):** AMS 310 (F'22, S'23), AMS 326 (S'24), AMS 510 (F'24; led recitations), AMS 595 (F'24), AMS 528 (S'25), AMS 502 (S'25); ran discussions, grading, and office hours.
 - **Instructor (Jan 2023 – Present):** AMS 394 (W'23, W'24, Su'24, Su'25), AMS 326 (Su'25), AMS 361 (W'26); designed/delivered lectures, created and graded assessments, and mentored students.
- **Courant Institute of Mathematical Sciences, New York University (NYU)** Fall 2020 – May 2022
Recitation Leader & Grader New York, NY, USA
 - **Recitation Leader (Jan 2021 – May 2022):** Mathematics for Economics II (Sp'21, Sp'22); Probability, Statistics & Decision Making (Fa'21, Sp'22). Led recitations, held office hours, and graded.
 - **Grader (Fall 2020, Fall 2021):** Analysis (Fa'20); Special Topics (Fa'21).

TALKS & PRESENTATIONS

- **Lightning Talk – Diffusion4PDE**, IACS Annual Meeting, 2025; **Lightning Talk – Active Learning for Neural Operators**, IACS Annual Meeting, 2024; **Student Seminar Speaker – AI for PDEs**, IACS Student Seminar, 2025.

ACADEMIC SERVICE

- Reviewer: *IJCNN* 2025/2026; *TNNLS* 2025; *ICLR* 2026; Gold Reviewer, *ICML* 2026; Reviewer, *NeurIPS* (formerly NIPS) 2026; *Neurocomputing* 2026.

REFERENCES

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